MIL-T-85472A(AS)
4 AUGUST 1987
SUPERSEDING
MIL-T-85472(AS)
14 May 1985

MILITARY SPECIFICATION

TEST SET, GROUP OQ-169B/DSM-96A, SPECIFICATION FOR

This specification is approved for use within the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the design, performance, and acceptance requirements for the Group OQ-169B/DSM-96A Test Set herein referred to as the test set. The test set is for shore Installation and is used with the Guided Weapon Test Set AN/DSM-96A to test the operational status of Walleye Data Link Weapons and Walleye Phase II Data Link Weapons.

2. APPLICABLE DOCUMENTS

2.1 Government Documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Systems Engineering and Standardization Department (SESD) Code 93, Lakehurst, NJ 08733 5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4935
Distribution Statement A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:54Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

FEDERAL

PPP-B-636	Boxes, Shipping, Fiberboard

MILITARY

MIL-P-116	Preservation and Packaging, Methods of
MIL-S-901	Shock Tests, H.I. (High Impact); Shipboard Machinery Equipment and Systems, Requirements for
MIL-T-5422	Testing, Environmental, Airborne Electronic and Associated Equipment
WS-6536	Procedures and Requirements for Preparation and Soldering of Electrical Connections
MIL-E-16400	Electronic, Interior Communication, and Navigation Equipment, Naval Ship and Shore, General Specification for
MIL-E-17555	Electronic and Electrical Equipment, Accessories, and Repair Parts; Packaging and Packing of
MIL-T-28800	Test Equipment for use with Electrical and Electronic Equipment, General Specification for
MIL-T-85031	Test Set, Guided Weapon, AN/DSM-96A

STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-704	Electric Power, Aircraft Characteristic and Utilization of
MIL-STD-781	Reliability Tests and Exponential Distribution
MIL-STD-810	Environmental Test Methods

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:54Z
Check the source to verify that this is the current version before use.

MIL-STD-965	Parts Control Program

2.1.2 Drawings and publications. The following drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

NAVAL AIR SYSTEMS COMMAND

DRAWINGS

DL412AS9000	Test Set, Group OQ-169B/DSM-96A

PUBLICATIONS

NAVAIR 16-350Q169-1	Maintenance Instructions with Illustrated Parts Breakdown (Intermediate-Depot) Test Set, Group OQ-169B/DSM-96A
NAVAIR 16-350Q169-1A	Secret Supplement, Test Set, Group OQ-169B/DSM-96A

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DODISS specified the supplement thereto, if applicable.

Instruction Manual for Power Meter and Thermistor Mount, Hewlett-Packard, Model HP-432A

Instruction Manual for Power Supply, Hewlett-Packard, Model HP-60155C

(Application for copies shall be addressed to the Hewlett-Packard Company, 1501 Page Mill Road, Palo Alto, California 94304.)

Instruction Manual for TV Monitor, Conrac Div, Giannini Control Corporation, Model PVA17/R or SNA17/R

(Application for copies shall be addressed to the Conrac Corp/Controls Group 3, Landmark Square, CT, 06901.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:54Z
Check the source to verify that this is the current version before use.

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Item description. Test Set Group OQ-169B/DSM-96A shall be used in conjunction with Guided Weapon Test Set AN/DSM-96A to operationally test the Walleye Data Link Guided Weapons. The test set shall consist of the principal assemblies shown on figure 1.

3.1.1 Interface description. The OQ-169B/DSM-96A Test Set Group shall be Interconnected with Guided Weapon Test Set AN/DSM-96A as shown on figure 2 for Walleye weapon testing. The OQ-169B and DSM-96A test consoles shall contain all the switches and indicators required to enter the test commands and monitor the responses of the weapon under test. The OQ-169B/DSM-96A Test Set Group shall be capable of checking all the parameters unique to data link/PHASE I and II weapons. All remaining parameters shall be checked with the DSM-96A test set (see MIL-T-85031). Figure 3 depicts a typical weapon test setup using OQ-169B and DSM-96A test sets as a unit.

3.2 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.3 Parts and materials.

3.3.1 Nonstandard parts and material approval. Approval for the use of nonstandard parts and materials shall be obtained by the contractor as specified in MIL-STD-965.

3.3.2 Basic conformance. Material selected for use in the equipment shall be as specified in MIL-T-28800, Parts shall be selected from those conforming to MIL-E-16400. unless the parts listed therein will not perform adequately in the intended applications and environments specified herein in which case 3.3.1 applies.

3.3.3 Interchangeability. All parts and materials shall meet the Interchangeability requirements as specified in MIL-T-28800. The drawings conforming to DL412AS9000 shall be the standard for measuring interchange-ability of all applicable units, and assembling subassemblies and parts.

3.4 Design and construction. The test set group shall be manufactured in accordance with the drawings and subsidiary specifications conforming to DL412AS9000, except that errors of omission or commission on the drawings may be incompatible with the requirements of this specification. In any case, it is a condition of final acceptance that the equipment manufactured shall meet all the design, construction, and workmanship requirements of this specification and applicable test documents.

3.4.1 Reliability.

3.4.1.1 Operational stability, The test set group shall operate with required performance, continuously or intermittently for a period of not less

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:54Z
Check the source to verify that this is the current version before use.

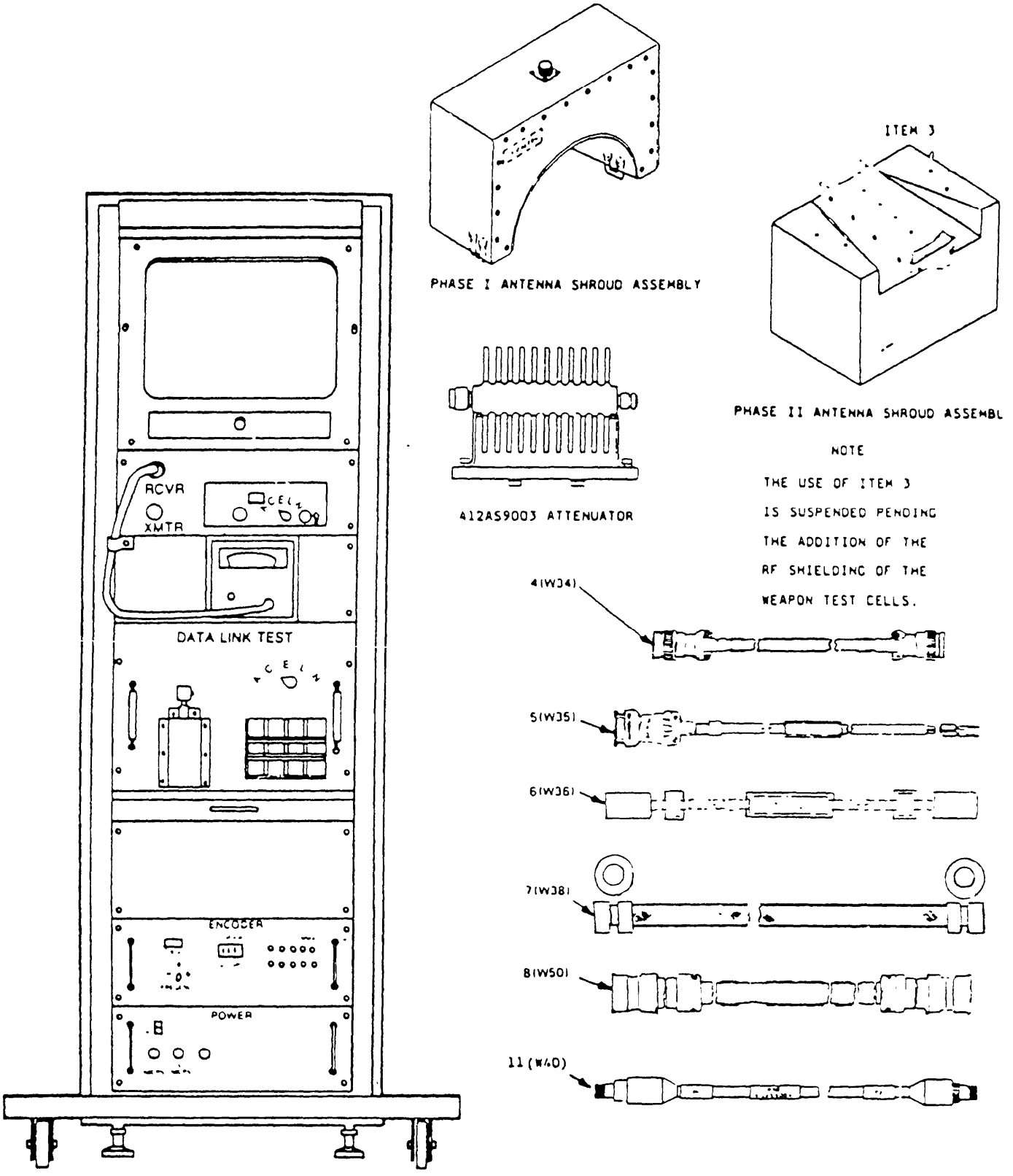


FIGURE 1. Test Set Group OQ-169B/DSM-96A.

Source: https://assist.dla.mil -- Downloaded: 202640-23T10:54Z
Check the source to verify that this is the current version before use.

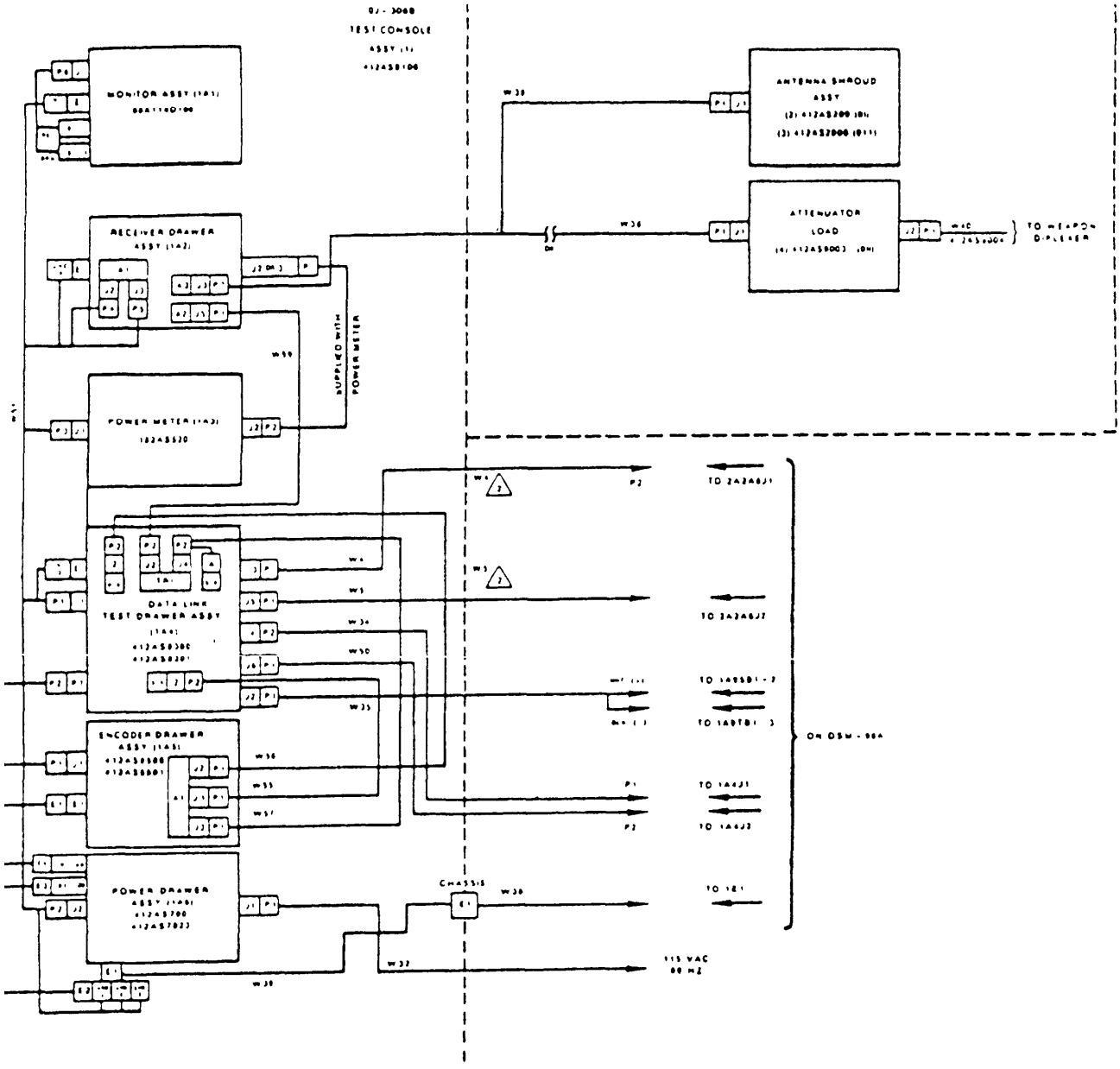


FIGURE 2. Test Set Group OQ-169B/DSM-96A interconnection diagram.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:54Z
Check the source to verify that this is the current version before use.

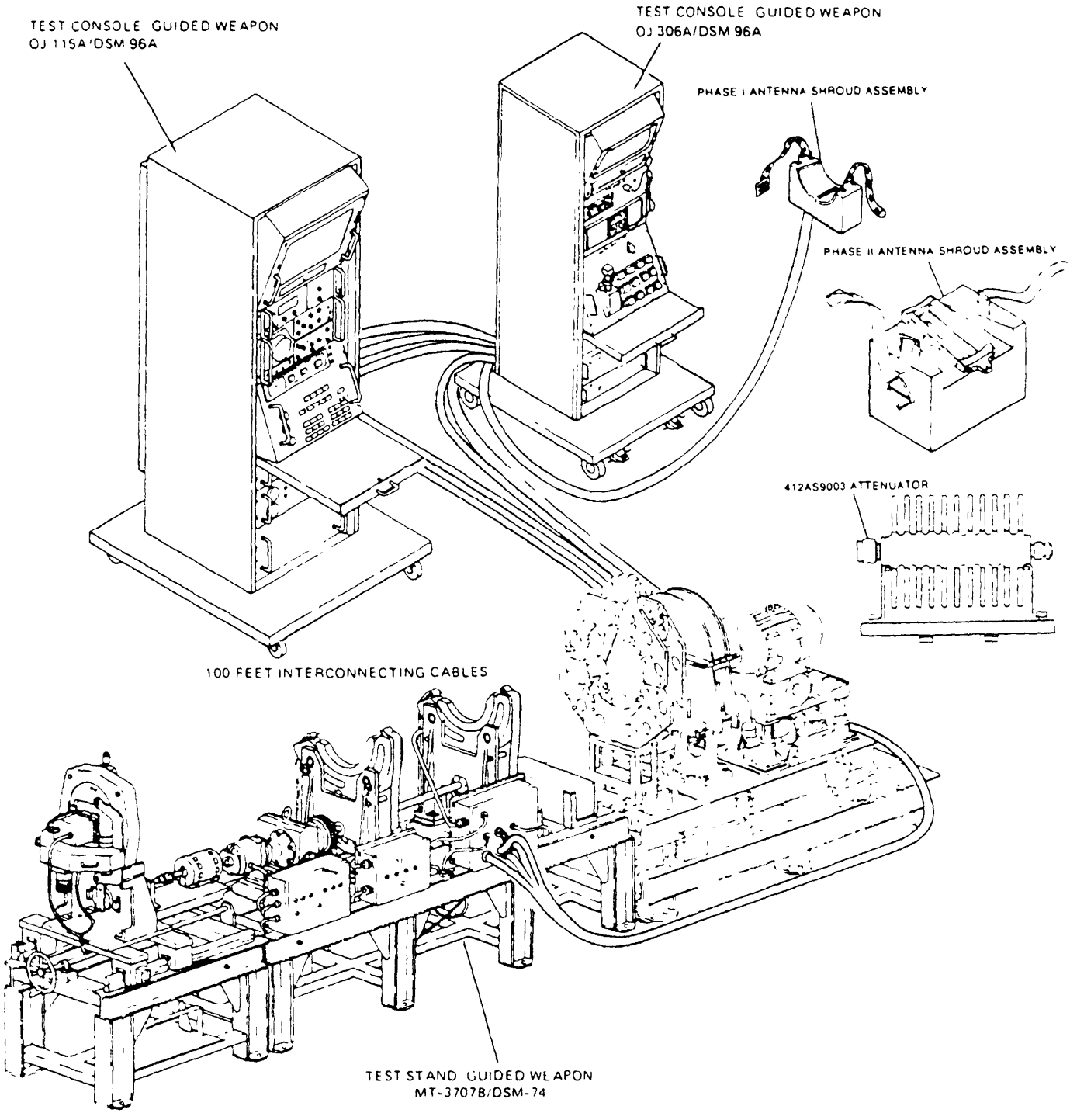


FIGURE 3. Typical Walleye test set ups.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:54Z
Check the source to verify that this is the current version before use.

than 300 hours. During this time, it shall not be necessary to readjust any controls that are inaccessible to the operator during normal use.

3.4.1.2 Operating life. The test set group shall have a total operating life of not less than 2000 hours with minimal servicing and replacement of parts.

3.4.1.3 Reliability in mean-time-between-failures (MTBF). The test set group shall have a specified mean (operating) time between failures of 300 hours when tested in accordance with 4.5.3.

3.4.2 Standard conditions. The following conditions shall be used to establish the normal performance characteristics under standard conditions for performing laboratory bench tests:

a.	Temperature	Room ambient (30 $\pm$10 ^{o}C)
b.	Altitude	Normal ground
c.	Vibration	None
d.	Humidity	Room ambient up to 90 percent relative humidity
e.	Input Power	115 $\pm$11.5 volts, 60 $\pm$3 Hz, 1 phase at 3400 volt-amperes, 30 (+2) (-0.0) volts dc, 5 amperes

3.4.2.1 Service conditions. The test set shall operate as required under any of the environmental service conditions or reasonable combination of these conditions as specified in MIL-E-16400. except as modified herein:

3.4.2.1.1 Temperature.

a. Operating. The test set group shall maintain specified performance while exposed to low operating temperatures of -O ^{o}C and high operating temperatures of +50 ^{o}C.

b. Non-operating. The test set group shall not be damaged nor shall operational performance be degraded when restored to the operational temperature range after being subjected to a nonoperating low temperature of -62 ^{o}C and a high nonoperating temperature of +71 ^{o}C.

3.4.2.1.2 Vibration. The test set group shall meet the requirements of 4.9.2 for equipment transported as secured cargo.

3.4.2.1.3 Shock. The test set shall be capable of withstanding the Grade A, Class I, shock requirements of MIL-S-901.

3.4.2.1.4 Humidity. The test set shall maintain the specified performance when exposed to a relative humidity of 95 percent for both continuous and Intermittent periods, including conditions wherein condensation takes place in and on the equipment in the form of both water and frost.

Source: https://assist.dla.mil -- Downloaded: 202640-23T10:54Z
Check the source to verify that this is the current version before use.

3.4.2.1.5 Fungus. The test set shall be resistant to fungi in both the operating and nonoperating condition and shall not be adversely affected when tested in accordance with 4.9.5.

3.4.3 Warm-up time. The time required for the equipment to warm-up prior to operation shall be not greater than 10 minutes under standard conditions and 30 minutes under service conditions.

3.4.4 Primary input power requirements. The test set shall meet input power requirements as specified in MIL-STD-704 for Class 2 equipment. The power required shall be of the following type and shall be not greater than the specified values:

a. 115 ±11.5 volts ac, 60 ±3 Hz, 1 phase, 3400 volt-amperes

b. 30 (+2.0) (-0.0) volts dc, 5 amperes

3.4.4.1 Electrical overload protection. Electrical overload protection for the test set group shall conform to Requirement 8 of MIL-STD-454 for Class 1 equipment.

3.4.5 Electromagnetic interference. The generation of electromagnetic interference by the test set and the susceptibility of the test set to electromagnetic interference shall be in accordance with MIL-STD-461 as referenced in MIL-E-16400.

3.4.5,1 Conducted susceptibility, power leads. The test set group shall not be susceptible to signals from 50 Hz to 400 MHz injected on the 115 V ac power leads as specified in MIL-STD-461, Requirement CS02.

3.4.5.2 Conducted emission, power leads. Conducted emissions from the test set group power leads shall be within the limits of MIL-STD-461, Requirements CE02 and CE04.

3.4.5.3 Conducted emission, control, and signal leads. Conducted emissions from the test set control and signal leads shall be within the limits of MIL-STD-46l, Requirements CE03 and CS05.

3.4.5.4 Radiated susceptibility. The test set shall not be susceptible to radiated signals from 30 Hz to 12.4 GHz as specified in MIL-STD-461, Requirements RS01, RS03 and RS03.1.

3.4.5.5 Radiated emission, electric field. Radiated emissions from the test set over the frequency range of 14 Hz to 1000 MHz, shall be within the limits of MIL-STD-461, Requirement RE02.

3.4.5.6 Transient voltage. The test set group shall meet the requirements of MIL-E-16400 for transient high and low voltage.

3.4.5.7 Transient frequency. The test set group shall meet the requirements of MIL-E-16400 for transient high and low frequency.

3.4.5.8 Power interruption. The test set group shall meet the requirements of MIL-E-16400 for power interruption.

3.4.6 Dimensions. The dimensions of the test set group shall be not greater than the following:

Source: https://assist.dla.mil -- Downloaded: 2026l0-23T10:54Z
Check the source to verify that this is the currenversion before use.

a. Height 78.76 inches

b. Width 40.5 inches

c. Depth 36.5 inches

3.4.7 Weight. The total weight of the test set group console shall be not greater than 700 pounds.

3.4.8 Color and finish. The color and finish of the test set shall be in accordance with DL412AS9000.

3.4.9 Identification plate and item markings. Nomenclature and identification marking shall be in accordance with MIL-STD-130 and DL412AS9000.

3.5 Performance. The performance of the test set group shall be as specified herein in addition to the performance requirements specified in DL412AS9000. Unless otherwise specified herein, these requirements shall apply under both standard (see 3.4.2) and extreme service conditions (see 3.4.2.1). When reduced performance requirements for service conditions are acceptable, such variations are specified herein.

3.5.1 Function and description. The function and description of the test set is covered in NAVAIR 16-350Q169-1. The test set shall provide the following functions (see figure 4):

a. Transmit azimuth (AZ) and elevation (EL) proportional guidance signals to weapon when in COMMAND OVERRIDE mode.

b. Transmit signals to control the weapon video transmitter on or off conditions, when commanded.

c. Monitor the video signal transmitted from the weapon.

d. Test control sections having a position gyro or rate gyro.

e. Monitor the power level of both transmitted and received RF signals.

3.5.2 Content. The test set shall consist of the following assemblies:

a. Test Console Assembly (1) PN 412AS9100 0J-306B/DSM-96A

b. Phase I Antenna Shroud Assembly (2A1) PN 412AS200

c. Attenuator PN 412AS9003

d. Phase II Antenna Shroud Assembly (3A1) PN 412AS2000

e. Cable Assembly (W32) PN 182AS3200

f. Cable Assembly (W34) PN 182AS3400

g. Cable Assembly (W35) PN 182AS3500

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:54Z
Check the source to verify that this is the current version before use.

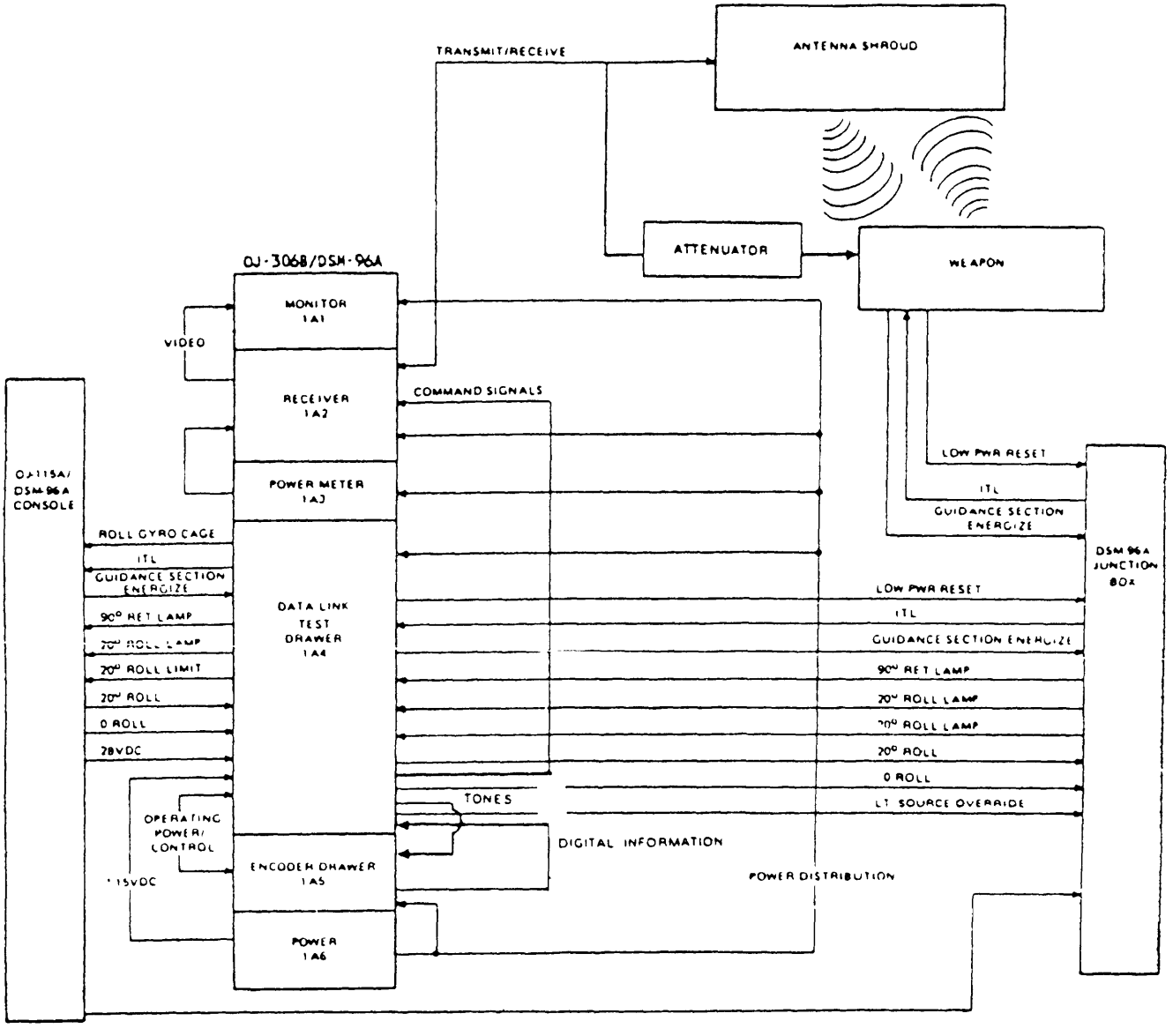


FIGURE 4. Test Set Group OO-169B/DSM-96A, block diagram.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:54Z
Check the source to verify that this is the current version before use.

h. Cable Assembly (W36) PN 182AS3600

i. Cable Assembly (W38) PN 182AS3800

j. Cable Assembly (W40) PN 412AS9004

k. Cable Assembly (W50) PN 412AS5000

3.6 Detail requirements.

3.6.1 Test Console Assembly, OJ-306B/DSM-96A. The test set console shall be manufactured in accordance with DL412AS9000 and shall meet the requirements stated thereon and in 3.6.1.1 and 3.6.1.2.

3.6.1.1 Function. The test console assembly shall have the capability of testing all the parameters unique to the data link and Phase 11 data link of Walleye weapons.

3.6.1.2 Console content. The test console assembly shall contain the following assemblies:

a. Monitor Drawer Assembly (1A1), PN 68A14D100

b. Receiver Drawer Assembly (1A2), PN 412AS8000

c. Power Meter Drawer Assembly (1A3), PN 412AS520

d. Data Link Test Drawer Assembly (1A4), PN 412AS8300

e. Encoder Drawer Assembly (1A5), PN 412AS8500

f. Power Drawer Assembly (1A6), PN 412AS700

g. Cabinet Assembly, PN 412AS3010

3.6.2 Monitor drawer assembly. The monitor drawer assembly shall be manufactured in accordance with DL412AS9000 and shall meet the requirements stated thereon and in 3.6.2.1 through 3.6.2.2.8.

3.6.2.1 Television (TV) Monitor Conrac Models PVA17/Rand SNA17/R. The principal functional element of the monitor drawer assembly shall be TV Monitor Conrac Model PVA17/R, SNA17/R or equal. For functional description of the TV monitor, refer to Instruction Manual for TV Monitor, Conrac Division, Giannini Corporation, Model PVA17/R or SNA17/R. The TV Monitor shall meet the requirements of the manufacturer's specification for TV Monitor Model PVA17/R or SNA17/R.

3.6.2.2 Monitor assembly function. The monitor assembly shall be capable of displaying the weapon video transmitter output and shall provide a means for visually-determining the quality of the target video and observing the response of the weapon to an override command signal (movement of cross-hair across target video). A panel located below the display tube shall provide the control and indicators for the monitor assembly.

3.6.2.2.1 ON switch. The ON switch shall apply 115 volts ac, single phase to the monitor.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:54Z
Check the source to verify that this is the current version before use.

3.6.2.2.2 CONTRAST control. The CONTRAST control shall vary the contrast of the display.

3.6.2.2.3 V LIN control. The V LIN control shall control the vertical linearity of the raster.

3.6.2.2.4 V SIZE control. The V SIZE control shall control the height of the raster.

3.6.2.2.5 V HOLD control. The V HOLD control shall stabilize the raster in the vertical plane.

3.6,2.2.6 BRIGHTNESS control. The BRIGHTNESS control shall vary the brightness of the display.

3.6.2.2.7 H HOLD control. The H HOLD control shall stabilize the raster in the horizontal plane.

3.6.2.2.8 INPUT Z switch. The INPUT Z switch shall provide the capability of selecting either HIGH or 75 ohms impedance for matching impedance of the monitor input to the input signal.

3.6.3 Receiver drawer assembly. The receiver assembly shall be manufactured in accordance with DL412AS9000 and shall meet the requirements stated thereon and in 3.6.3.1 through 3.6.3.1.7.

3.6.3.1 Receiver assembly functions. The receiver assembly shall be capable of receiving the video signal from the weapon under test and demodulating the signal for display on the monitor (1A1). The front panel of the receiver drawer assembly shall contain coaxial jacks, J2 (RCVR) and J3 (XMTR), and the operator controls for the receiver.

3.6.3.1.1 RCVR J2 connector. Connector J2 shall provide a connection for power meter 1A3 for measuring RF power level of weapon transmitter.

3.6.3.1.2 XMTR J3 connector. Connector J3 shall provide a connection for power meter 1A3 to test set transmitter 1A4 TR1 for measurement of RF power level.

3.6.3.1.3 RF channel control. The RF channel control shall provide the capability of selecting any one of the five receiver center frequencies.

3.6.3.1.4 Power switch. The power switch shall apply the 115 volt ac, 60 Hz single phase voltage to the receiver assembly.

3.6.3.1.5 Meter function switch. The meter function switch shall select the signal strength or tuning.

3.6.3.1.6 Video level control. Video level control shall adjust the output video level.

3.6.3.1.7 Video output jack. The video output jack shall provide a monitor connection for video level during calibration and weapon testing.

3.6.4 Power meter drawer assembly. The power meter assembly shall be manufactured in accordance with DL412AS9000 and shall meet the requirements stated thereon and in 3.6.4.1 through 3.6.4.2.7.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:54Z
Check the source to verify that this is the current version before use.

3.6,4.1 Power Meter Model HP-432A. The principal functional element of the power meter drawer assembly shall be Power Meter Model HP-432A or equal. For functional description of the power meter, refer to Instruction Manual for Power Meter and Thermistor Mount, Hewlett-Packard Model HP-432A. The power meter shall meet the manufacturer's specification for Power Meter Model HP-432A.

3.6.4.2 Power meter function. The power meter shall be capable of measuring the RF power of the weapon video received and telemetry signals from the test set transmitter. The self lead coaxial cable of the power meter shall be capable of being connected to receiver drawer connectors J2 (for measuring video) and J3 (for measuring telemetry). The front panel of the power meter shall contain the connector, controls and indicators for the power meter.

3.6.4.2.1 Thermistor mount connector. The thermistor mount connector shall connect the thermistor mount to the power meter.

3.6.4.2.2 Meter mechanical zero control. The meter mechanical zero control shall provide capability for zero adjusting the meter using a screwdriver.

3.6.4.2.3 POWER ON-OFF switch. This switch shall control the 115 volt ac, 60 Hz power to the power meter. The AC LINE indicator shall illuminate when POWER ON-OFF switch is in the ON position indicating power is applied to the power meter.

3.6.4.2.4 mw-RANGE-dBm switch. The mw-RANGE-dBm switch shall provide capability for the selection of proper meter range.

3.6.4.2.5 COARSE ZERO control. The COARSE ZERO control shall provide capability for the meter to be roughly zero adjusted, electrically.

3,6.4.2.6 FINE ZERO control. The FINE ZERO control shall provide capability for the meter to be more accurately adjusted, electrically.

3.6.4.2.7 CALIBRATION FACTOR switch. The CALIBRATION FACTOR switch shall provide capability for selecting the desired calibration factor.

3.6.5 Data link test assembly. The data link test assembly shall be manufactured in accordance with DL412AS9000 and shall meet the requirements stated thereon and in 3.6.5.1 through 3.6.5.1.21.

3.6.5.1 Data link test assembly functions. The data link test assembly shall be the central point for inserting data link and telemetry commands (see figure 5). Data link switches/controls shall connect the outputs of voltage controlled oscillators to the transmitter via mixer amplifier for data link commands or via mixer amplifier and encoder assembly for data link Phase II commands. The data link test assembly controls and major subassemblies shall be as specified in 3.6.5.1.1 through 3.6.5.1.2.1.

3.6.5.1.1 CONTROLLER (CT1). The CONTROLLER (Joy Stick) shall be capable of remote controlling the weapon's target tracking circuit. The COMMAND OVERRIDE switch 1A4S11, mounted on the CONTROLLER, shall provide the capability for overriding the weapon's automatic tracking circuits allowing the CONTROLLER to function. The controller shall consist of two pairs of semiconductor strain gauges mounted on a beam assembly in the controller. Pressure applied to the controller shall provide two dc outputs to a

Source: https://assist.dla.mil -- Downloaded: 202640-23T10:54Z
Check the source to verify that this is the current version before use.

controller amplifier 1A4A1. The magnitude and polarity of the dc voltage outputs from the strain gauges shall depend on the direction and amount of force applied to the controller. When pressed, the command override switch 1A4S11 shall energize relay 1A4K5 which shall, in turn:

a. activate the controller amplifier

b. apply the output of the 10.5 KHz VCO to the mixer 1A4A2

c. apply +28 volts dc to COMMAND OVERRIDE indicator 1A4DS2

3.6.5.1.1.1 Controller amplifier 1A4A1. The controller amplifier 1A4A1 shall be a dc amplifier used to amplify the dc output of the controller strain gauges. The controller amplifier shall have two identical channels, one for azimuth (AZ) and one for elevation (EL). The dc output of each shall be applied to the input of the appropriate VCO (EL-30 KHz, AZ-22 KHz). The output shall also be applied through contacts of deenergized relay 1A4K5 to the integrating reference amplifier (part of controller amplifier). The integrating reference amplifier output shall be used to zero the dc amplifier output when not in command override mode. When COMMAND OVERRIDE 1A4S11 switch is pressed, relay 1A4K5 shall energize and interrupt the feedback path to the integrating reference amplifier. Resistor R15 (EL) and R29 (AZ) shall be adjusted for a null at cathode of CR1 and cathode of CR3 respectively when not in command override mode. Power required for the controller amplifier is $\pm$15 volts dc. Positive and negative 15 volts dc shall be supplied to the strain gauges from the controller amplifier. The two proportional 22 KHz and 30 KHz channels shall be frequency deviated by the dc output voltage of the controller amplifier 1A4A1.

3.6.5.1.2 DATA LINK/STANDARD switch (1A4S1). The DATA LINK/STANDARD switch shall provide the capability for testing data link weapons when in the DATA LINK switch position or shall deactivate the test set for standard Walleye weapon testing when in the STANDARD switch position. In the DATA LINK switch position, relay 1A4K3 shall be energized applying !15V dc and +28V dc to the data link test drawer. Also the roll gyro cage circuit shall be opened and a special circuit (including relays K1, K2, and K6) shall be activated for the roll portion of the torque test.

3.6.5.1.3 G/S ON - G/S OFF switch (1A4S3). This two position switch shall interrupt the +28V dc guidance section energize command, to the weapon, from the AN/DSM-96A Test Console.

3.6.5.1.4 TEST ON - TEST OFF (1A4S3). This two position switch shall provide some of the logic for selecting weapon video power HI or LO by connecting 1.7 KHz VCO output to mixer 1A4A2, when in the TEST ON position.

3,6.5.1.5 XMTR ON - XMTR OFF switch (1A4S4). This two position switch shall apply +28V dc energizing power to the test set transmitter 1A4TR1 in the XMTR ON position.

3.6.5.1.6 VIDEO ON switch (1A4S5). This momentary type switch shall connect the 5.4 KHz VCO output to the mixer 1A4A2 and shall apply +28V dc to latching relay 1A4K8 when switch is pressed. Relay 1A4K8, in turn, shall connect +28V dc to VIDEO ON indicator 14AS5.

3.6.5.1.7 VIDEO OFF switch (1A4S6). This momentary type switch shall connect the 7.35 KHz VCO output to mixer 1A4A2 and shall apply +28V dc to

Source: https://assist.dla.mil -- Downloaded: 202610-23T10:54Z
Check the source to verify that this is the current version before use.

latching relay 1A4A2 when switch is pressed. Relay 1A4K8, reset, shall apply +28V dc to VIDEO OFF indicator 1A4S6 and shall remove the +28V dc from VIDEO ON indicator.

3.6.5.1.8 DL A/DL B switch (1A4S7). This two position switch shall select the proper circuit for correct amount of weapon roll during the torque tests. DL A switch position shall select the 25° roll, and DL B switch position shall select the 45° roll.

3.6.5.1.9 COMMAND OVERRIDE switch (1A4CT1S11). When pressed, the COMMANDOVERRIDE switch shall energize relay 1A4K5 which shall energize the control amplifier 1A4A1. The control amplifier shall apply the 10,5 KHz VCO output to mixer 1A4A2 and +28V dc shall be applied to the COMMAND OVERRIDE indicator 1A4DS2. The 10.5 KHz signal shall override the weapon's automatic tracking circuits to permit the CONTROLLER to function.

3.6.5.1.10 OVERRIDE/NORMAL switch (1A4S13). The OVERRIDE/NORMAL switch shall select the high tracking rate instead of the normal tracking rate during torque tests. (The higher tracking rate for the data link weapon is necessary to adequately check the yaw and pitch circuits.)

3.6.5.1.11 2.3 KHZ switch (1A4S12). The 2.3 KHz switch, when pressed, shall activate PSI flag by connecting the 2.3 KHz VCO output to mixer 1A4A2.

3.6.5.1.12 ITL OPEN/ITL CLOSE switch (1A4S14). THE ITL OPEN/ITL CLOSE switch shall interrupt the intent-to-launch (ITL) line between the AN/DSM-96A console and the weapon when switch is pressed and held.

3.6.5.1.13 FREQUENCY select switch 1A4S10. The FREQUENCY select switch shall be d five position rotary switch for selecting any one of the five center frequencies of the test transmitter 1A4TR1. The five frequencies are classified and are in classified supplement Technical Manual NAVAIR 16-350Q169-1A.

3.6.5.1.14 VCO TEST - 22 KHz control. The VCO TEST - 22 KHz control (S8) shall provide a test function for adjusting the output level of the 22 KHz voltage controlled oscillator.

3.6.5.1.15 VCO TEST - 30 KHz control. The VCO TEST - 30 KHZ control (S9) shall provide a test function for adjusting the output level of the 30 KHz voltage controlled oscillator.

3.6.5.1.16 25° roll test circuit. The 25° roll test circuit, consisting of relays 1A4K1, K2 and K6, shall cause the weapon to roll 25° instead of 45° that-the AN/DSM-96A test set is programmed for when testing large scale weapons.

3.6.5.1.17 20° roll test circuit. When the DATA LINK/STANDARD switch 1A4K1 is in DATA LINK position and the DL A/DL B switch 1A4S7 is in the DL A position, a special 20° roll circuit shall be enabled. This circuit shall be activated when the roll command (+28V dc from the AN/DSM-96A) is applied to relay 1A4K2 and the 20° roll relay 2A2A7K1 (in the MT-3707B/DSM-74 Guided Weapon Test Stand).

3.6.5.1.18 Voltage control oscillator mount (1A4VCOMI). The oscillator mount has connections for 10 plug-in VCOs. There shall presently be eight VCOs installed to provide the various command signals to be transmitted.

Source: https://assist.dla.mil -- Downloaded: 20260-23T10:54Z
Check the source to verify that this is the current version before use.

Six of the VCOs shall be operated at fixed frequencies with the outputs being applied to the mixer 1A4A2 when the appropriate switch on the data test panel is pressed. Table I lists the commands and associated VCO frequencies.

3.6.5.1.19 Mixer amplifier (1A4A2). The amplifier shall be a basic operational amplifier capable of summing the VCO outputs. The amplifier shall be capable of summing 10 inputs but only eight shall be presently used. Again the potentiometer in the feedback path shall allow the composite modulation signal amplitude to be adjusted to the proper level.

TABLE I. Test set commands and associated VCO frequencies.

Command	VCO Frequency
Pod Antenna Select	1.3 KHz
Test On	1.7 KHz
PSI Flag	2.3 KHz
Video On	5.4 KHz
Video Off	7.35 KHz
Command Override	10.5 KHz
AZ Proportional Command	22 KHz
EL Proportional Command	30 KHz

The mixer output shall be connected to coaxial switch 1A4S15 for switching to the transmitter 1A4TR1 or the Encoder Drawer Assembly 1A5. The mixer shall be energized with ±15V dc.

3.6.5.1.20 Transmitter (1A4TR1). The transmitter shall be a five channel VHF FM transmitter. The transmitter shall be frequency modulated by the composite signal from the mixer amplifier 1A4A2. The transmitter shall require +28V dc for heater supply and +28V dc for transmitter power on. The +28V dc for the heater supply shall be applied when DATA LINK is selected, and the +28V dc power on is provided through the XMTR ON/XMTR OFF switch 1A4S4.

3.6.5.1.21 Coaxial switches (1A4S15 and 1A4S16). These coaxial switches shall be activated by the TONE/DIGITAL switch 1A5S14 (in the Encoder Drawer 1A5). With switch 1A5S14 in the TONE position, the mixer output shall be directed through 1A4S15 and 1A4S16 to the transmitter modulation input. With 1A5S14 in the DIGITAL position, coaxial switch 1A4S15 shall direct the mixer output to the Encoder Drawer 1A5 via cable W55.

3.6.6 Encoder drawer assembly. The encoder assembly shall be manufactured in accordance with DL412AS9000 and shall meet the requirements stated thereon and in 3.6.6.1 through 3.6.6.1.6.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:54Z
Check the source to verify that this is the current version before use.

3.6.6.1 Encoder functions. The encoder drawer shall be capable of converting the mixed data link command signal from the data link test drawer, and various discrete control signals to a Phase II Serial output. The mixed signal shall be applied to the encoder drawer via cable W55 (see figure 2). The mixed signals shall then be converted into a binary code and fed into a series of multiplexer. The output of the multiplexer shall be converted to Phase II format. The telemetry information for the transmitter in the data link test drawer shall be supplied via cable W56 and shall be modulated by the Phase 11 signal from the half-sine wave modulator 1A5A6.

3.6.6.1.1 FREQUENCY switch. The FREQUENCY switch shall be a three-position rotary switch which shall provide the capabilities for selecting a positive or negative dc voltage to change the carrier frequency of the transmitter via cable W57.

3.6.6.1.2 TONE - DIGITAL switches/indicators. These switches/indicators shall be a two position push-button switch/indicator which shall provide the capability to activate the encoder drawer in the Phase II mode of operation.

3.6.6.1.3 Phase II CODE switches. These switches shall be eight digit thumbwheel switches which shall provide the capability to enter the code number onto the Phase 11 signal for transmission to the weapon.

3.6.6. 1.4 DISCRETE COMMAND (S1 through S10) switches. These switches shall be OFF/ON toggle switches which shall provide the capability to enter commands into the Phase II signal for transmission to the weapon.

3.6.6.1.5 Phase II format conversion. Signals from the Mixer Amplifier (1A4A2) shall be converted to Phase II format by the Encoder Assembly 1A5A1 in the Encoder Drawer Assembly. The Encoder Assembly shall decode and convert the Mixer Amplifier signals to a bit rate word. The Phase II signal shall then be routed to the data-link Test Drawer (1A4) to modulate the transmitter 1A4TR1, and in turn to weapon under test. The bit rate word format is classified. Refer to SECRET manual NAVAIR 16-350Q169-1A for the word format.

3.6.6.1.6 TONE/DIGITAL switch (1A5S14). With (TONE/DIGITAL) switch 1A5S14 in TONE position, the output of the transmitter 1A4TR1 shall be directed to the Phase I weapon under test. With switch 1A5S14 in the DIGITAL position, the output of the 1A4TR1 transmitter shall be directed to the Phase II weapon under test.

3.6.6.2 Encoder assembly (1A5A1). The encoder assembly shall consist of six printed wiring boards 1A5A1A1 through 1A5A1A6 housed in a card cage assembly.

3.6.6.2.1 Printed wiring boards (1A5A1A1 and 1A5A1A2). These printed wiring boards shall decode five discrete commands (Test On, PSI Flag, Video On, Video Off, and Command Override) from the mixer amplifier 1A4A2 in the Data Link Test Drawer (1A4).

3.6.6.2.2 Printed wiring board (1A5A1A3). A bit rate control signal shall be generated on printed wiring board 1A5A1A3. This bit rate signal shall be supplied to the clock input of a seven stage ripple counter. The output of the counter shall be gated to provide the timing control signals to the multiplexer and counters. The ripple counter shall reset after every word. To count the azimuth and elevation frequencies, bandpass filters shall be

Source: https://assist.dla.mil -- Downloaded: 202610-23T10:54Z
Check the source to verify that this is the current version before use.

used on board 1A5A1A3 to remove unwanted positions of the tone multiplexed signal. Filter 1A5A1A3 FL1 shall be used for the 22 KHz azimuth signal and filter 1A5A1A3 FL2 shall be used for the 30 KHz elevation signal. Comparators shall be used to convert the filter outputs to logic level compatible square waves.

3.6.6.2.3 Printed wiring board (1A5A1A4). Counting of the azimuth and elevation frequencies shall be performed on the 1A5A1A4 board. These guidance signals shall be counted by preset binary up/down counters (preset to 177 for azimuth and 168 for elevation) and shall be gated every 10 clock pulses (azimuth) or 12 clock pulses (elevation). This binary output then shall become part of the Phase II data stream. In the Phase II data stream, a logic level one shall be noted by a change of phase during the period. While there is no change for a logic 0, there shall always be a change of phase at the transition from one bit period to the next. This shall be accomplished on the 1A5A1A4 board by a J-K Flip Flop and two NOR gates combining the data stream and the bit rate signal.

3.6.6.2.4 Printed wiring board (1A5A1A5). To assure the weapon will receive a command, it shall be necessary for the command to be transmitted for a minimum of 3 seconds. A multivibrator on the 1A5A1A3 board shall be used to produce a 3 second minimum transmission time for the VIDEO ON and VIDEO OFF commands. The 3-second delay to the remaining command signals shall be generated by two quad timers on the 1A5A1A5 board.

3.6.6.2.5 Printed wiring board (1A5A1A6). The circuitry on the 1A5A1A6 board shall supply an adjustable + or - dc voltage to offset the transmitter carrier. These dc voltages shall be selected by the FREQUENCY select switch 1A5S13 on the front panel and passed on to the transmitter via cable W57. The 1A5A1A6 board also converts the binary Phase II data stream into pulses to modulate the transmitter 1A4TR1 for phase II operation.

3.6.7 Power drawer assembly. The power drawer assembly shall be manufactured in accordance with DL412AS9000 and shall meet the requirements stated thereon and specified in 3.6.7.1 through 3.6.7.2.3.

3.6.7.1 Power Supply Model HP-60155C. The principal functional element of the power drawer assembly shall be Power Supply Model HP-60155C or equal. For functional description of the power supply, refer to Instruction Manual for Power Supply, Hewlett-Packard, Model HP-60155C. The power supply shall meet the requirements of the manufacturer's specification for Power Supply HP-60155C.

3.6.7.2 Power functions. The power drawer assembly shall be capable of controlling all the power circuitry except the 28V dc which shall be supplied to the data link test drawer from the AN/DSM-96A Test Console. Externally supplied 115 volt, 60 Hz single phase power shall be applied to all of the test set assembles except the data link test drawer, via cable assemblies W51 and W53 (see figure 2). The 115V ac shall also be applied to the 115V dc power supply in the power drawer. The output of the !15V dc power supply shall be applied to the data link test drawer for command signal generation via cable assembly W51. The front panel shall contain the controls and indicators for the power drawer.

3.6.7.2.1 ON/OFF switch (S1). This switch shall be a toggle switch which shall control the 115V, 60 Hz single phase power applied to the test console.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:54Z
Check the source to verify that this is the current version before use.

3.6.7.2.2 115V, 60 CPS indicator. This indicator lamp shall illuminate when 115V, 60 Hz power is applied to the test console.

3.6.7.2.3 5A, 115V, 60 CPS fuse (F1). This 5-ampere fuse shall provide circuit protection for 115V, 60 Hz power.

3.6.8 Cabinet console assembly. The cabinet console assembly shall be manufactured in accordance with DL412AS9000 and shall meet the requirements stated thereon and as specified in 3.6.8.1.

3.6.8.1 Cabinet basic requirements. The cabinet shall be manufactured of steel and shall be capable of supporting five rack-mounted assemblies, a front mounted assembly and a pull-out desk. The cabinet shall be supported on two fixed and two swivel casters. A lift-out access panel shall be provided for the back of the cabinet permitting easy access to the rear of the drawer assemblies.

3.6.9 Phase I antenna shroud assembly (2A1). The Phase I antenna shroud shall be manufactured in accordance with DL412AS9000 and shall meet the requirements stated thereon and specified in 3.6.9.1.

3.6.9.1 Phase I antenna shroud description and functions. The shroud assembly shall consist of a transmit/receive antenna enclosed in a shroud. The shroud antenna shall be capable of receiving from and radiating to the weapon antenna. The shroud shall be capable of preventing radiated power of the antenna from exposing nearby equipment and personnel. This shroud shall be designed for use in testing Phase I simulated data link weapons.

3.6.10 Phase II antenna shroud assembly (3A1). The Phase II antenna shroud shall be manufactured in accordance with DL412AS9000 and shall meet the requirements stated thereon and specified in 3.6.10.1.

3.6.10.1 Phase II antenna shroud description and functions. The shroud assembly shall consist of a transmit/receive antenna enclosed in a shroud. The shroud antenna shall be capable of receiving and radiating signals to and from the weapon antenna. The shroud shall be capable of preventing this radiated power of the antennas from exposing nearby equipment and personnel, This shroud shall be designed for use in testing Phase II simulated data link weapons.

3.6.11 Cable assemblies. Cable assemblies shall be manufactured to meet the requirements specified in 3.6.11.1 through 3.6.11.7.

3.6.11.1 Cable assembly (W32). This cable assembly shall be used for 115V ac, 60 Hz power cable. The cable assembly shall be manufactured in accordance with DL412AS9000.

3.6.11.2 Cable assembly (W34). This cable assembly shall be manufactured in accordance with DL412AS9000.

3.6.11.3 Cable assembly (W35). This cable assembly shall be manufactured in accordance with DL412AS9000.

3.6.11.4 Cable assembly (W36). This cable assembly shall be manufactured in accordance with DL412AS9000.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:54Z
Check the source to verify that this is the current version before use.

3.6.11.5 Cable assembly (W38). This cable assembly shall be manufactured in accordance with DL412AS9000.

3.6.11.6 Cable assembly (W40). This cable assembly shall be manufactured in accordance with DL412AS9000.

3.6.11.7 Cable assembly (W50). This cable assembly shall be manufactured in accordance with DL412AS9000.

3.7 Soldering. Preparation, procedures and requirements for soldering of electrical connections shall be in accordance with WS-6536.

3.8 Workmanship. The test set group, including all parts and accessories, shall be manufactured in accordance with DL412AS9000 and as specified herein. Workmanship shall meet the requirements of MIL-STD-454, Requirement 9.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification, where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)

b. Quality conformance inspection (see 4.5)

4.3 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the conditions specified as follows:

a. Temperature	Room ambient (25 $\pm$10 ^{o}C)
b. Altitude	Normal ground
c. Vibration	None

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:54Z
Check the source to verify that this is the current version before use.

d. Humidity	Room ambient up to 90 percent relative humidity
e. Input Power	115 ±11.5 volts ac, 60 ±3 Hz 30 (+2, -0) volts dc

4.3.1 Performance characteristics. The test set shall satisfy the requirements for interference control (see 3.3.5) and warm-up time (see 3.3.3) before testing. During inspection acceptance testing the equipment shall satisfy the requirements of Section 3.

4.4 First article inspection. First article tests shall be made on a test set representative of the production test sets to be supplied under the contract. First article test performance shall be the responsibility of the contractor and shall be performed in accordance with approved test procedures (see 4.14). No first article inspections shall be conducted prior to approval of the first article test procedure by the procuring activity.

4.4.1 Scope of first article tests. First article tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification and the contract. These shall include environmental and interference tests in accordance with MIL-T-5422 as specified in MIL-T-28800. Test methods shall be in accordance with MIL-STD-810 except as modified by this specification.

4.4.2 First article approval. Approval of the first article shall be done by the procuring activity upon satisfactory completion of all tests.

4.4.3 Production test sets. Test sets supplied under the contract shall, in all respects including design, construction, workmanship, performance and quality, be equal to the approved first article sample. Each test set shall be capable of successfully completing the same test as imposed on the first article sample.

4.5 Quality conformance inspection. The contractor shall furnish all samples and shall be responsible for accomplishing the quality conformance tests. All inspection and testing shall be to the approved quality conformance test procedures under the supervision of the Government inspector.

a. Individual test (see 4.5.1)

b. Sampling test (see 4.5.2)

c. Special tests (see 4.5.4)

4.5.1 Individual tests. Each test set submitted for acceptance shall be subjected to the individual tests. Those tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each test set accepted shall have passed the following tests:

a. Examination of product	(see 4.5.1.1)
b. Operational test	(see 4.5.1.2)
c. Manufacturing run-in test	(see 4.5.1.3)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:54Z
Check the source to verify that this is the current version before use.

4.5.1.1 Examination of product. Each test set shall be examined carefully to determine that the material and workmanship requirements have been met.

4.5.1.2 Operational test. Each test set shall be operated long enough to permit the temperature to stabilize and to check sufficient characteristics and record adequate data to assure required equipment operation.

4.5.1.3 Manufacturing run-in test. Each test set shall be operated under the following conditions for a period of 6 hours without failure. A failure shall be considered as anything which causes malfunctioning of the test set. Only those adjustments will be permitted which can be made using controls that are accessible to the operator during normal use of the equipment.

a. Temperature	Room ambient
b. Humidity	Room ambient
c. Vibration	Any selected frequency within range of 20 to 30 Hz (excluding resonant points) and a minimum amplitude of ± 3 g's.

The test set shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 6-hour period of operation, Where feasible, the test set shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration shall be vertical to the normal mounting plane for 5 minutes and lateral to the plane for 5 minutes. Where it is not feasible to vibrate the test set in two directions, the vertical shall be used. During the 6-hour period of operation following the 10-minute vibration period, the test set shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, the test set shall be repaired and the test started over, except that the 10-minute vibration period need not be repeated when it is established that the failure was not a result of vibration, Should failures occur repeatedly, corrective action shall be taken to eliminate this defect from future test sets. The 6-hour period specified above may be made up of several shorter periods to conform to normal working hours.

4.5.2 Sampling tests. Test sets selected for sampling tests shall first have passed the individual tests. Test sets will be selected for sampling tests by the Government inspector in accordance with table II.

TABLE II. Sampling test selection.

Quantity of equipments offered for acceptance	Quantity to be selected for sampling tests
First 5	1
Next 10	1
Next 25	1
Next 50	1
Next 100	1
	1 for each additional 200 or fraction thereof.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:54Z
Check the source to verify that this is the current version before use.

4.5.2.1 Scope of tests. As a minimum, each test set selected for sampling tests shall be subjected to the following:

a. Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

b. Operational test at certain environmental conditions. The conditions may vary for each test set tested and should be based on results of the first article, individual, and special tests.

c. Manufacturing run-in test specified in 4.5.1.3 except that the test duration shall be 120 hours with no restrictions on the number of failures.

4.5.3 Reliability assurance tests. Reliability assurance tests shall be conducted in accordance with the requirements on MIL-STD-781. Tests shall be conducted as required for both the qualification phase and the production acceptance phase.

4.5.3.1 Reliability qualification phase. Prior to acceptance of test sets under contract or purchase order, a contractually specified quantity of test sets (at least one if not specified) shall be subjected to the combined reliability qualification requirements for fixed ground test set (category 1) of MIL-STD-781, using Test Plan IC therein (θ_1 = 200) (see 6.2.1).

4.5.3.2 Reliability acceptance phase. Production lot samples of the test sets shall be subjected to the combined production reliability acceptance requirements for fixed group equipment (category 1) of MIL-STD-781, using Test Plan VC therein (θ_1 = 100).

4.5.3.2.1 Acceptance phase sample size. The entire contract quantity (except for those test sets used in the qualification phase) shall be considered one lot. The reliability acceptance sample size shall be ten percent of the contract quantity (at least one unit). The test program shall start the first month after the qualification phase has been completed. The sample(s) shall be tested until an accept/reject decision has been reached. In the meantime, other test sets manufactured may be shipped in place, if acceptable to the procuring officer. The procuring activity reserves the right to stop the acceptance of test sets anytime after one or more reject decisions have been reached, pending a review of the contractor's efforts to improve the test sets, parts, and quality control in order that the entire contract quantity will result in an accept decision.

4.5.3.3 Test details. The test details shall be part of test procedures prior to the beginning of the qualification test phase of the reliability assurance tests.

4.5.3.3.1 Temperature extremes. The temperature extremes shall be limited to -62 °C to +71 °C storage and -O °C to +50 °C operating.

4.5.3.3.2 Voltage cycling. Voltage cycling in accordance with MIL-STD-781 is not required.

4,5.4 Special tests. Special tests shall be conducted on a quantity of test sets as specified in the contract (see 6.2.1) for the purpose of checking the

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:54Z
Check the source to verify that this is the current version before use.

effect of any design or material change on the performance of the test set and to assure quality control. The test sets selected for special tests may be from test sets previously subjected to the sampling tests.

4.5.4.1 Special test schedule. Selection of test sets for special tests shall be made as follows:

a. On early test sets after an engineering or material change.

b. Whenever failure reports or other evidences indicate that additional tests are required. (This will be determined by the procuring activity.)

4.5.4.2 Scope of tests. Special tests shall consist of such tests acceptable to the procuring activity. Test procedures previously approved for the first article tests may be used where applicable.

4.6 Equipment failure. Should a failure occur during the sampling, reliability assurance or special tests, the following action shall be taken:

a. Determine the cause of failure.

b. Determine if the failure is an isolated case or due to design defect.

c. Where practical, include a test (in the individual test) to check all test sets for this requirement until assurance is forthcoming that the defect has been corrected.

4.7 Life test. The contractor shall furnish all samples and shall be responsible for accomplishing the life test. The life test shall be of 300 hours duration and shall be conducted on test sets that have passed the individual tests. The life test shall be performed under the conditions specified in 4.7.1. The life test samples shall be selected by the Government inspector in accordance with table III. (Test sets which have successfully completed the sampling tests or special tests may be selected for life tests.) The life test may be omitted, if accumulated operating time during other testing for a quantity of test sets is equal to or more than that listed in table III after having received at least 300 hours each of test time.

TABLE III. Life test sample selection.

Quantity of test sets offered for acceptance	Quantity to be selected for life test
First 1	1
Next 10	1
Next 100	1
	One for each additional 100 or fraction thereof.

4,7.1 Life test conditions. The life test shall be conducted under the following simulated service conditions:

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:54Z
Check the source to verify that this is the current version before use.

a. Temperature	Room ambient (30 °C ±10 °C)
b. Altitude	Normal ground (0 - 5000 ft.) (0 - 1600 meters)
c. Humidity	Room ambient up to 90 percent relative humidity
d. Input Voltage	Nominal
e. Vibration	None

4.7.2 Life test period. The life test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the test set shall be turned on and off several times, and put through its various phases of operation.

4.7.3 Performance check. At approximately 8-hour intervals during the test, a limited performance check shall be made.

4.7.4 Test data. In the event of part failures, the defective part shall be replaced and the operation resumed for the remainder of the test period.

4.7.4.1 Failure report. In the event of a failure, the Government inspector shall be notified immediately.

4.8 Presubmission testing. No item, part, or complete test set shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply with all applicable requirements of this specification and the contract.

4.9 Environmental tests. The test set shall operate as required when subjected to the tests specified in 4.9.1 through 4.9.1.2 (see 3.4.2.1).

4.9.1 Temperature. The operating and nonoperating temperature test sequence shall be as specified in 4.9.1.1 and 4.9.1.2.

4.9.1.1 Low temperature. The test set group shall be subjected to low temperature tests in accordance with Method 502 of MIL-STD-810, except as follows:

a. Step 2 - Storage or nonoperating temperature shall be -62 °C (+0 °C) (-5 °C) for 24 hours.

b. Step 4 - Lowest operating temperature shall be 0 °C.

c. The combined duration of Steps 4 and 5 shall be not less than 12 hours, at least 2 of which shall be in the stabilized condition prior to conducting tests.

4.9.1.2 High temperature. The test set group shall be subjected to high temperature tests in accordance with Method 501, Procedure I, of MIL-STD-810, except as follows:

a. Step 4 - The highest operating temperature shall be +50 °C.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:54Z
Check the source to verify that this is the current version before use.

b. The combined duration of Steps 4 and 5 shall not be less than 12 hours, at least 6 of which shall be in the stabilized condition prior to conducting tests.

c* Step 7 - The test set shall be operated for a period of at least 2 hours in the stabilized condition before the Step 7 tests are performed.

4.9.2 Vibration. The test set group shall be subjected to vibration tests in accordance with Method 514.2, Procedure X, of MIL-STD-810. The equipment shall not be energized during vibration exposure of MIL-S-901.

4.9.3 Shock. The test set group shall be subjected to the Grade A, Type A, Class I, shock test of MIL-S-901.

4.9.4 Humidity. The test set group shall be humidity tested in accordance with Method 507.1, Procedure IV, of MIL-STD-810, except as follows:

a. Step 5 - The test set performance measurements shall be made at +50 ^{o}C.

b. Step 9 - The test set shall be carefully examined to detect evidence of physical degradation. Blistering or organic coatings shall be cause for rejection.

4.9.5 Fungus. The test set group shall be subjected to a fungus test in accordance with Method 508 of MIL-STD-810.

4.10 Warm-up time. After being energized 10 minutes, verify that the test set group meets the operational requirements of 4.5.1.2.

4.11 Primary input power tests. The test set shall be subjected to Steady State Voltage and Frequency Tests of MIL-E-16400.

4.11.1 Electrical overload protection test. The test set shall be tested for electrical overload protection in accordance with Requirement 8 of MIL-STD-454 for Class 1 equipment.

4.12 Electromagnetic interference (EMI) tests. The following EMI tests shall be performed in accordance with MIL-STD-462.

4.12.1 Conducted susceptibility, power leads. This test shall be performed in accordance with MIL-STD-462, Method CS02.

4.12.2 Conducted -emission, power leads. This test shall be performed in accordance with MIL-STD-462, Methods CE02 and CE04.

4.12.3 Conducted emission, control and signal leads. This test shall be performed in accordance with MIL-STD-462, Methods CE03 and CE05.

4.12.4 Radiated susceptibility. This test shall be performed in accordance with MIL-STD-462, Methods RS01, RS02, and RS03.

4.12.5 Radiated emission. This test shall be performed in accordance with MIL-STD-462, Method RE02.

Source: https://assist.dla.mil -- Downloaded: 2026-04-23T10:54Z
Check the source to verify that this is the current version before use.

4.12.6 Transient voltage. This test shall be conducted in accordance with MIL-E-16400. Post test data shall be in accordance with 4.5.1.2.

4.12.7 Transient frequency. This test shall be conducted in accordance with MIL-E-16400. Post test data shall be in accordance with 4.5.1.2.

4.12.8 Power interruption. This test shall be conducted in accordance with MIL-E-16400. Post test data shall be in accordance with 4.5.1.2.

4.13 Packaging inspection. Prior to shipment, inspect the test set packaging to determine conformance to the requirements of Section 5.

4.14 Methods of examination and test. The right is reserved by the procuring activity or the Government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification and the contract.

5. PACKAGING.

5.1 General. Packaging shall be in accordance with the general requirements of MIL-E-17555, supplemented as specified in 5.2 and 5.3.

5.2 Preservation. Unless otherwise specified in the contract or purchase order, preservation shall be in accordance with MIL-P-116, Method 111, Level A (see 6.2.1).

5.3 Packing. Unless otherwise specified in the contract or purchase order, packing shall be Level C (see 6.2.1). The packaged test sets shall be packed in fiberboard containers conforming to PPP-B-636, class weather resistant, style optional, normal requirements. Sealing (taping) and reinforcing (banding) shall be in accordance with the appendix to PPP-B-636.

5.4 Marking. In addition to any special marking specified in the contract (see 6.2.1), each unit package, intermediate and exterior container, and unitized load shall be marked in accordance with MIL-STD-129.

6. NOTES,

6.1 Intended use. The Test Set Group OQ-169B/DSM-96A is intended for use with Guided Weapon Test Set AN/DSM-96A to effect a complete check of the operational status of all Walleye data link guided weapons. The test set group is intended to be used in depot and intermediate maintenance activities to test assembled weapons and to aid in fault isolation of failed weapons.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification

b. Selection of applicable levels of packaging, packing, and preservation (see 5.2, 5.3)

c. If special marking is required (see 5.4)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:54Z
Check the source to verify that this is the current version before use.

d. When a first article is required for inspection and approval (see 4.3 and 6.3)

e. When special tests are to be performed (see 4.5.4)

f. If test procedures require modification (see 4.14)

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

Paragraph No.	Data requirement title	Applicable DID no.	Option
4.4, 4.14	Procedure, First Article Inspection	DI-T-4901	
4.4.1, 4.4.3	Report, First Article Inspection	DI-T-4902	
4.5.4.2, 4.7.3, 4.14	Procedures, Production/Acceptance Inspection	DI-T-4903	
4.5	Reports, Production Inspection	DI-T-4904	
4.5.3.2.1	Reports, Reliability Test and Demonstration	DI-R-7034	
4.5.3.3	Procedures, Reliability Tests	DI-R-7035	
4.6c	Plan, Failure Data Collection, Analysis and Corrective Action	UDI-T-23719	

(Data item descriptions related to this specification, and identified in section 6, will be approved and listed as such in DoD 5010.12-L, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 First article. When a first article inspection is required, the item should be a first article sample. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first article. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

Source: https://assist.dla.mil -- Downloaded: 202610-23T10:54Z
Check the source to verify that this is the current version before use.

6.4 Miscellaneous notes.

6.4.1 Performance objectives. Minimum size and weights, simplicity of operation, ease of maintenance, and improvement in the performance and reliability of the specific functions beyond the requirements of the specification are objectives which should be considered in the production of this equipment. Where it appears that substantial reduction in size or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in DL412AS9000, it is desired that their use be investigated. When investigation shows advantages can be realized, a request for approval should be submitted to the procuring activity for consideration. Each request should be accompanied by complete supporting information.

6.5 Abbreviations. Abbreviations used in this specification are listed as follows:

a.	AZ	Azimuth
b.	EL	Elevation
c.	FSK	Frequency shift keyed
d.	ITL	Intent to launch
e.	PSI	Pod status indicator
f.	MTBF	Mean-time-between-failures
g.	VCO	Voltage controlled oscillator

6.6 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.7 Subject term (keyword) Listing.

Antenna
Antenna shroud
Data link weapons
Guided weapon
Target video
Television monitor
Test set
Video
Walleye

Preparing Activity:
Navy-AS
(Project 4935-N066)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:54Z
Check the source to verify that this is the current version before use.

APPENDIX

INPUT/OUTPUT PARAMETERS

10. SCOPE

10.1 Scope. This appendix provides the input/output parameters for the Group OQ-169B/DSM-96A Test Set. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

20. APPLICABLE DOCUMENTS

This section is not applicable to this appendix.

30. REQUIREMENTS

30.1 Requirements. Input/output parameters for the test set are specified in tables IV and V. Voltage controlled oscillators output characteristics are specified in table VI. A calibration checklist for the test set is provided in table VII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:54Z
Check the source to verify that this is the current version before use.

TABLE IV. Phase I data link modulated system.

Command Link (Test set to weapon)	Test Set OQ-169B/DSM-96A		Walleye Weapon MK-21	
1. Frequency Tolerance	----- See Supplement NAVAIR 16-350Q169-1A ----- f_o ± 0.004%		Rx will track f_o ± 0.006%	
2. Signal Level	-86.5 ± 6.5 dBm Min Test Set Power Level to WPN RX		-92 dBm Min WPN RX Sensitivity	
3. VCO's (FM System)	Frequency	P-P Deviation	Frequency	P-P Deviation
	30 ± .060 KHz	102.86 KHz ±5143 KHz	30 ± .900 KHz	102.86 KHz ±7200.0
	22 ± .050	61.72 ±3086	22 ± .660	61.72 ±4320
	10.5 ± .030	24.00 ±1200	10.5 ± .315	24.00 ±1680
	7.35 ± .020	15.40 ± 770	7.35 ± .220	15.40 ±1078
	5.4 ± .020	10.28 ± 514	5.4 ± .162	10.28 ± 720
	2.3 ± .020	8.58 ± 429	2.3 ± .069	8.58 ± 601
	1.7 ± .010	8.58 ± 429	1.7 ± .051	8.58 ± 601
	1.3 ± .010	8.58 ± 429	1.3 ± .039	8.58 ± 601
	Deviation that the test set provides		Deviation that the WPN RX will handle	
	TOTAL TOL	240.00 KHz 5%	TOTAL TOL	240.00 KHz 7%
Video Link (Weapon to test set)				
1. Frequency Tolerance	----- See Supplement NAVAIR 16-350Q169-1A ----- Rx f_o ± .06%		Tx f_o ± .03%	
2. Signal Level	-5.85 dBm (min at pwr mtr) approx. -36 dBm at Rx		+41.16 dBm Min from WPN TX to WPN Ant	

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:54Z
Check the source to verify that this is the current version before use.

APPENDIX

TABLE V. Phase II data link modulated system.

Command Link (Test set to weapon)	Test Set OQ-169B/DSM-96A	Walleye Weapon MK-29
1. Frequency Tolerance	----- See Supplement NAVAIR 16-350Q169-1A ----- $f_o \pm 0.004\%$	Classified
2. Signal Level	-87 ± 7 dBm Min test set power Level to WPN RX	-97 dBm Min WPN RX Sensitivity
3. VCO's (FM System)	N/A	VCO Signals are Encoded to Phase II Format by Test Set — N/A
4. Phase II System Format Clock Rate Word Length Bit Assignments	----- See Supplement NAVAIR 16-350Q169-1A -----	
Video Link (Wea to test set)		
1. Frequency Tolerance	----- See Supplement NAVAIR 16-350Q169-1A ----- Rx $f_o \pm .06\%$	Tx $f_o \pm .03\%$
2. Signal Level	-3.6 dBm (min at pwr mtr) approx. -34 dBm at Rx	+43.42 dBm (min frm WPN TX to wpn ant)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:54Z
Check the source to verify that this is the current version before use.

APPENDIX

TABLE VI. Voltage controlled oscillators.

VCO	Nominal	Tolerances
1	0.036 VAC 1.700 KHz	0.031 to 0.042 VAC 1.690 to 1.710 KHz
2	0.036 VAC 2.300 KHz	0.031 to 0.042 VAC 2.280 to 2.320 KHz
3	0.040 VAC 5.400 KHz	0.036 to 0.051 VAC 5.380 to 5.420 KHz
4	0.065 VAC 7.350 KHz	0.056 to 0.071 VAC 7.330 to 7.370 KHz
5	0.100 VAC 10.50 KHz	0.089 to 0.107 VAC 10.47 to 10.53 KHz
6	0.255 VAC 22.000 KHz	0.240 to 0.269 VAC 21.950 to 22.050 KHz
7	0.412 VAC 30.000 KHz	0.403 to 0.450 VAC 29.940 to 30.060 KHz
8	0.036 VAC 1.300 KHz	0.031 to 0.042 VAC 1.290 to 1.310 KHz
CONTROLLER FORWARD WITH C/O DEPRESSED	27.75 KHz	27.35 to 28.15 KHz
CONTROLLER AFT WITH C/O DEPRESSED	32.25 KHz	31.85 to 32.65 KHz
CONTROLLER LEFT WITH C/O DEPRESSED	23.65 KHz	23.25 to 24.05 KHz
CONTROLLER RIGHT WITH C/O DEPRESSED	20.35 KHz	19.95 to 20.75 KHz

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:54Z
Check the source to verify that this is the current version before use.

APPENDIX

TABLE VII. Calibration Checklist for OO-169B/DSM-96A Test Set Group.

CALIBRATION CHECKLIST

TEST INST(S): 00169B/DSM-96A TEST SET GROUP

PROC. NO. MFR MODEL SERIAL NO.

PROCEDURE STEP NO. (1)	FUNCTION TESTED (2)	NOMINAL (3)	MEASURED VALUES 1st RUN (4)	2nd (5)	OUT of TOL (6)	CAL. TOL. (7)
4.1	VOLTAGE CONTROLLED OSCILLATORS					
4.1.2	AMPLITUDE:					VAC
4.1.2.1	CH 6, VCO 1 E21	0.036				+0.031 TO +0.042
4.1.2.2	CH 7, VCO 2 E39	0.036				+0.031 TO +0.042
4.1.2.3	CH 10, VCO 3 E23	0.044				+0.036 TO +0.051
4.1.2.4	CH 11, VCO 4 E25	0.064				+0.056 TO +0.071
4.1.2.5	CH 12, VCO 5 E27	0.098				+0.089 TO +0.107
4.1.2.6	CH 14, VCO 6 E29	0.255				+0.240 TO +0.269
4.1.2.7	CH 15, VCO 7 E31	0.427				+0.403 TO +0.450
4.1.2.8	CH 5, VCO 8 E41	0.036				+0.031 TO +0.042
4.1.3	FREQUENCY					KHZ
4.1.3.3	CH 6, VCO 1 TP	1.7 KHz				1.690 TO 1.710
4.1.3.4	CH 7, VCO 2 TP	2.3 KHz				2.280 TO 2.320
4.1.3.5	CH 10, VCO 3 TP	5.4 KHz				5.380 TO 5.420
4.1.3.6	CH 11, VCO 4 TP	7.35 KHz				7.330 TO 7.370
4.1.3.7	CH 12, VCO 5 TP	10.5 KHz				10.470 TO 10.530
4.1.3.8	CH 14, VCO 6 TP	22 KHz				21.950 TO 22.050
4.1.3.9	CH 15, VCO 7 TP	30 KHz				29.940 TO 30.060
4.1.3.10	CH 5, VCO 8 TP	1.3 KHz				1.290 TO 1.310

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:54Z
Check the source to verify that this is the current version before use.

TABLE VII. Calibration Checklist for OO-169B/DSM-96A Test Set Group - Continued.

CALIBRATION CHECKLIST

TEST INST(S): 00169B/DSM-96A TEST SET GROUP

PROC. NO.	MFR	MODEL	SERIAL NO.			
PROCEDURE STEP NO. (1)	FUNCTION TESTED (2)	NOMINAL (3)	MEASURED VALUES 1st RUN (4)	2nd (5)	OUT of TOL (6)	CAL. TOL. (7)
4.1.4.1	Controller Left	23.65 KHz				23.25 TO 24.05
4.1.4.2	Controller Right	20.35 KHz				19.95 TO 20.75
4.1.4.3	Controller FWD	27.75 KHz				27.35 TO 28.15
4.1.4.4	Controller AFT	32.25 KHz				31.85 TO 32.65
4.2 RECEIVER:						
4.2.1 CENTER FREQUENCY TEST*						
4.2.1.5	F_o A	150 KHz				149.838-150.163
4.2.1.6	F_o A	150 KHz				149.837-150.164
4.2.1.6	F_o E	150 KHz				149.835-150.165
4.2.1.6	F_o L	150 KHz				149.832-150.168
4.2.1.6	F_o N	150 KHz				149.830-150.170
4.2.3	Sensitivity Test					
4.2.3.6	Channel A	>= -70dB				
4.2.3.7	Channel C	>= -70dB				
4.2.3.7	Channel E	>= -70dB				
4.2.3.7	Channel L	>= -70dB				
4.2.3.7	Channel N	>= -70dB				

* Refer to NAVAIR 16-350Q169-1A

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:54Z
Check the source to verify that this is the current version before use.

APPENDIX

TABLE VII. Calibration Checklist for OO-169B/DSM-96A Test Set Group - Continued.

CALIBRATION CHECKLIST

TEST INST(S): 00169B/DSM-96A TEST SET GROUP

PROC. NO.	MFR	MODEL	SERIAL NO.

PROCEDURE STEP NO. (1)	FUNCTION TESTED (2)	NOMINAL (3)	MEASURED VALUES 1st RUN (4)	2nd (5)	OUT of TOL (6)	CAL. TOL. (7)
4.2.4	Bandwidth Test					
4.2.4.8	Channel A	10.5 MHz				7 TO 14 MHz
4.2.4.9	Channel C	10.5 MHz				7 TO 14 MHz
4.2.4.9	Channel E	10.5 MHz				7 TO 14 MHz
4.2.4.9	Channel L	10.5 MHz				7 TO 14 MHz
4.2.4.9	Channel N	10.5 MHz				7 TO 14 MHz
4.3	TRANSMITTER					
4.3.1	POWER OUTPUT					
4.3.1.6	CH A Tone	-10.5 dBm				-8 TO -11 dBm
4.3.1.6	CH C Tone	-10.5 dBm				-8 TO -11 dBm
4.3.1.6	CH E Tone	-10.5 dBm				-8 TO -11 dBm
4.3.1.6	CH L Tone	-10.5 dBm				-8 TO -11 dBm
4.3.1.6	CH N Tone	-10.5 dBm				-8 TO -11 dBm
4.3.1.7	CH A DIGITAL	-10.5 dBm				-8 TO -11 dBm
4.3.1.7	CH C DIGITAL	-10.5 dBm				-8 TO -11 dBm
4.3.1.7	CH E DIGITAL	-10.5 dBm				-8 TO -11 dBm
4.3.1.7	CH L DIGITAL	-10.5 dBm				-8 TO -11 dBm
4.3.1.7	CH N DIGITAL	-10.5 dBm				-8 TO -11 dBm

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:54Z
Check the source to verify that this is the current version before use.

TABLE VII. Calibration Checklist for OQ-169B/DSM-96A Test Set Group - Continued.

CALIBRATION CHECKLIST

TEST INST(S): 00169B/DSM-96A TEST SET GROUP

PROC. NO.	MFR	MODEL	SERIAL NO.

PROCEDURE STEP NO. (1)	FUNCTION TESTED (2)	NOMINAL (3)	MEASURED VALUES 1st RUN (4)	2nd (5)	OUT of TOL (6)	CAL. TOL. (7)
4.3.2	CENTER FREQUENCY*					
4.3.2.3	CH A					+/-0.004%
4.3.2.3	CH C					+/-0.004%
4.3.2.3	CH E					+/-0.004%
4.3.2.3	CH L					+/-0.004%
4.3.2.3	CH N					+/-0.004%
4.3.3	INCIDENTAL FREQUENCY MODULATION TEST					
4.3.3.4	CH A					<= 10 KHZ
4.3.3.5	CH C					<= 10 KHz
4.3.3.5	CH E					<= 10 KHz
4.3.3.5	CH L					<= 10 KHz
4.3.3.5	CH N					<= 10 KHz
4.3.4	CARRIER NULL					
4.3.4.11	1A4X12-2					
4.4	ENCODER ASSEMBLY					
4.4.6	A1 MODULE PIN 6	5V P-P				+/-0.2 V P-P
4.4.7	A1 MODULE PIN 8	220 mvP-P				+/-20 mv P-P
4.4.11	A6 MODULE PIN 6	180 mvP-P				+/-10 mv P-P

* Refer to NAVAIR 16-350Q169-1A

Source: https://assist.dla.mil -- Downloaded: 202610-23T10:54Z
Check the source to verify that this is the current version before use.

TABLE VII. Calibration Checklist for OQ-169B/DSM-96A Test Set Group - Continued.

CALIBRATION CHECKLIST

TEST INST(S): 00169B/DSM-96A TEST SET GROUP

PROC. NO. MFR MODEL SERIAL NO.

PROCEDURE STEP NO. (1)	FUNCTION TESTED (2)	NOMINAL (3)	MEASURED VALUES 1st RUN (4)	2nd (5)	OUT of TOL (6)	CAL. TOL. (7)
4.5 TRANSMITTER FREQUENCY OFFSET						
4.5.7	FREQ SW (-)	50 KHz				48 TO 52 KHz
4.5.9	FREQ SW (+)	50 KHz				48 TO 52 KHz
4.6 INSERTION LOSS MEASUREMENTS FOR DIPLEXERS, ATTENUATORS, DIRECTIONAL COUPLERS, AND COAXIAL, CABLES. (RF PATHS)						
4.6.1 COMPONENT GROUP A INSERTION LOSS						
4.6.1.8	CH A	1.5 dB				0 TO 3 dB
4.6.1.8	CH C	1.5 dB				0 TO 3 dB
4.6.1.8	CH E	1.5 dB				0 TO 3 dB
4.6.1.8	CH L	1.5 dB				0 TO 3 dB
4.6.1.8	CH N	1.5 dB				0 TO 3 dB
4.6.2 COMPONENT GROUP B INSERTION LOSS						
4.6.2.8	CH A	31 dB				29 TO 33 dB
4.6.2.8	CH C	31 dB				29 TO 33 dB
4.6.2.8	CH E	31 dB				29 TO 33 dB
4.6.2.8	CH L	31 dB				29 TO 33 dB
4.6.2.8	CH N	31 dB				29 TO 33 dB
4.6.3 COMPONENT GROUP C INSERTION LOSS						
4.6.3.7	CH A	22 dB				20 TO 24 dB
4.6.3.7	CH C	22 dB				20 TO 24 dB

Source: https://assist.dla.mil -- Downloaded: 202610-23T10:54Z
Check the source to verify that this is the current version before use.

TABLE VII. Calibration Checklist for OO-169B/DSM-96A Test Set Group - Continued.

CALIBRATION CHECKLIST

TEST INST(S): 00169B/DSM-96A TEST SET GROUP

PROC. NO. | MFR | MODEL | SERIAL NO.

PROCEDURE STEP NO. (1)	FUNCTION TESTED (2)	NOMINAL (3)	MEASURED VALUES 1st RUN (4)	2nd (5)	OUT of TOL (6)	CAL. TOL. (7)
4.6.3.7	CH E	22 dB				20 TO 24 dB
4.6.3.7	CH L	22 dB				20 TO 24 dB
4.6.3.7	CH N	22 dB				20 TO 24 dB
4.6.4	COMPONENT GROUP D INSERTION LOSS					
4.6.4.8	CH A	68.5 dB				65 TO 72 dB
4.6.4.9	CH C	68.5 dB				65 TO 72 dB
4.6.4.9	CH E	68.5 dB				65 TO 72 dB
4.6.4.9	CH L	68.5 dB				65 TO 72 dB
4.6.4.9	CH N	68.5 dB				65 TO 72 dB
4.6.5	COMPONENT GROUP E INSERTION LOSS					
4.6.5.5	CH A	42.5 dB				39 TO 46 dB
4.6.6.6	CH C	42.5 dB				39 TO 46 dB
4.6.6.6	CH E	42.5 dB				39 TO 46 dB
4.6.6.6	CH L	42.5 dB				39 TO 46 dB
4.6.6.6	CH N	42.5 dB				39 TO 46 dB

Source: https://assist.dla.mil -- Downloaded: 202610-23T10:54Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE NAVY
Commanding Officer
Naval Air Engineering Center
Systems Engineering Standardization Department
(SESD), Code 93
Lakehurst, NJ 08733

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO 12503 WASHINGTON D C

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

Commanding Officer
Naval Air Engineering Center
Systems Engineering Standardization Department
(SESD). Code 93
Lakehurst, NJ 08733

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:54Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-85472A	TEST SET, GROUP QQ-169B/DSM-96A SPECIFICATION FOR

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)* _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:54Z
Check the source to verify that this is the current version before use.